

Accessibility in Astronomy

HTML on arXiv, alt text for figures, and conferences

How do we
make
astronomy
more
accessible and
inclusive to
people with
disabilities?

- HTML on arXiv
 - Why it's great!
 - How to use it
- Alternative text
 - What is it?
 - How to write it
- A couple of thoughts on accessibility at conferences

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um overabundances and oxygen
axy. To address the scarcity of these
a classification accuracy of

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HTML is a new paper format available on arXiv (for recent papers) which is much more accessible than LaTeX-generated PDFs.

Why do we need an HTML format?

- A bit of background...
 - Screen readers: read all text content on the screen and allow navigation without any visuals
 - VoiceOver on MacOS and iOS; NVDA and JAWS on Windows; Orca on Linux
 - Screen magnifiers: magnify content on the screen
 - Text-to-speech: reads written text aloud

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 - Text-to-speech: reads written text aloud
- Many different assistive technologies exist because many disabilities exist—and most disabilities (including visual impairment) exist on a spectrum.

The speaking rate is lowered from my usual preferred speed for easier listening!

DRAFT VERSION FEBRUARY 6, 2025
Typeset using L^AT_EX twocolumn style in AASTeX631

Radial migration in the Galactic disc driven by a slowing bar

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ABSTRACT

Radial migration is an important dynamical effect that has reshaped the Galactic disc, but its origin has yet to be elucidated. In this work, we present evidence that resonant dragging by the corotation of a decelerating bar could be the main driver of radial migration in the Milky Way disc. Using a test particle simulation, we demonstrate this scenario explains the two distinct age-metallicity sequences observed in the solar vicinity: the plateauing upper sequence is interpreted as stars dragged outwards by the expanding corotation of the decelerating bar and the steeper lower sequence as stars formed locally around the solar circle. The upper migrated sequence dominates at guiding radii around the current corotation radius of the bar, $R \sim 7$ kpc, but rapidly dies away beyond this where the mechanism cannot operate. This behaviour naturally explains the radial dependence of the $[\alpha/\text{Fe}]$ -bimodality, in particular the truncation of the high- $[\alpha/\text{Fe}]$ disc beyond the solar circle. Under our proposed radial migration scenario, we constrain the Milky Way bar's pattern speed evolution using the age-metallicity distribution of stars currently trapped at corotation. We find the bar likely formed

2v1 [astro-ph.GA] 4 Feb 2025



Abstract

1 Introduction

2 Data

3 Simulation

4 Results

5 Discussion

6 Conclusion

References

slowing-down history of the Galactic bar from it.

4.3.1 A motivational calculation

In a realistic galaxy model, the metallicities of stars depend on their birth time and radius, i.e. $[\text{Fe}/\text{H}](t, R)$. Hence, the age-metallicity gradient of a sequence can be written as

$$\frac{D[\text{Fe}/\text{H}]}{Dt} = \frac{\partial[\text{Fe}/\text{H}]}{\partial R} \frac{dR}{dt} + \frac{\partial[\text{Fe}/\text{H}]}{\partial t}. \quad (2)$$

Under the scenario that the migrated (upper) age-metallicity sequence is a consequence of the dragging of the bar corotation resonance, we relate dR/dt to the bar deceleration rate as

$$\frac{dR_{\text{CR}}}{dt} = -\frac{\dot{\Omega}}{\Omega^2} V_c = \eta V_c. \quad (3)$$

We then rearrange Eq. (2) as

$$\eta = \frac{\partial[\text{Fe}/\text{H}]/\partial t - D[\text{Fe}/\text{H}]/Dt}{|\partial[\text{Fe}/\text{H}]/\partial R|} \frac{1}{V_c}. \quad (4)$$

We can estimate $\partial[\text{Fe}/\text{H}]/\partial t$ from the slope of the local (lower) age-metallicity sequence in Fig. 1, which yields $\partial[\text{Fe}/\text{H}]/\partial t \approx 0.075 \text{ dex Gyr}^{-1}$. We adopt $V_c \approx 230 \text{ km s}^{-1}$ and $\partial[\text{Fe}/\text{H}]/\partial R \approx -0.07 \text{ dex/kpc}$ (McMillan, 2017; Bragança et al., 2019). As the age-metallicity gradient of the migrated sequence is almost flat, $D[\text{Fe}/\text{H}]/Dt \approx 0$, we calculate $\eta \approx 0.0046$ from Eq. (4). From Eq. (2), we can see that the flatness of the migrated age-metallicity sequence is a consequence of the balance between the bar deceleration rate and the metallicity enrichment rate. A decline or increase in the migrated age-metallicity sequence after bar formation could also be expected if the deceleration rate is faster or slower.

× . Hence, the age-metallicity gradient of a sequence can be written as with the birth radii of stars

radial migration in the Galactic disc in greater detail with inferred birth radii of stars. With stellar metal-
measurements, Lu et al. (2024b) improved on the method described in Minchev et al. (2018) and derived
the empirical metallicity temporal evolution, $[\text{Fe}/\text{H}](R_b, \tau)$, with the same subgiant star sample that we used in this work
(Xiang & Rix, 2022). Using the metallicity evolution, they calculated the birth radii of these stars. The methodology is
tested against various suites of cosmological simulations (see Lu et al. 2023; Batcliff et al. 2024b; Lu et al. 2024a, b

Report Issue

HTML is useful for low vision accessibility, too!

Way under the bar-driven radial migration scenario.

2 Data

We use the stellar parameter and isochrone age measurements of $\sim 247\,000$ subgiant stars derived from LAMOST DR7 spectra by *Xiang et al. (2019)* and *Xiang & Rix (2022)*. The typical age uncertainty is $\sim 7.5\%$ for stars younger than 10 Gyr and $\sim 10\%$ for older stars. To obtain stellar kinematics, we use the astrometric measurements from Gaia DR3 (*Gaia Collaboration et al., 2022*), geometric distances derived in *Bailer-Jones et al. (2021)*, and the line-of-sight velocity measurements from LAMOST (*Deng et al., 2012*). We calculate the orbital parameters (e.g. eccentricity and guiding radius) of stars using the Milky Way potential in *McMillan (2017)* using AGAMA (*Vasiliev, 2019*). To ensure reliable orbital parameters, we select stars with small parallax uncertainties and small radial velocity uncertainties, i.e. $\varpi/\sigma_{\varpi} > 5$ and $\sigma_{v_{\text{los}}} < 10 \text{ km s}^{-1}$. We remove stars with metallicity error $\sigma_{[\text{Fe}/\text{H}]} > 0.1$ or age error $\sigma_{\tau} > 1 \text{ Gyr}$. To select stars belonging to the Galactic disc, we keep

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Radial migration in the Galactic disc driven by a slowing bar

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Radial migration is an important dynamical effect that has reshaped the Galactic disc, but its origin has yet to be elucidated. In this work, we present evidence that moment dragging by the concretion of a decelerating bar could be the main driver of radial migration in the Milky Way disc. Using a two-particle simulation, we demonstrate this scenario explains the two distinct age-metallicity sequences observed in the solar vicinity: the plateauing upper sequence is interpreted as stars dragged outwards by the expanding concretion of the decelerating bar and the steeper lower sequence as stars formed locally around the solar circle. The upper internal sequence dominates at galactic radii around the current concretion radius of the bar, $R \approx 7$ kpc, but rapidly dies away beyond this where the mechanism cannot operate. This behaviour naturally explains the radial dependence of the [α]/[Fe] bimodality, in particular the truncation of the high-[α]/[Fe] disc beyond the solar circle. Under our proposed radial migration scenario, we constrain the Milky Way bar's pattern speed evolution using the age-metallicity distribution of stars currently trapped at concretion. We find the bar likely formed with an initial pattern speed of $60 - 100$ km s $^{-1}$ kpc $^{-1}$ and began decelerating 6–8 Gyr ago at a rate $-0.312 \pm 0.0425 - 0.0346$ before the present epoch (which includes systematic uncertainties).

Keywords: Milky Way dynamics(1053) – Galactic bar(2365) – Milky Way evolution(1052) – Milky Way disk(1050) – Galaxy evolution(594)

1. INTRODUCTION

Stars in the galactic disc can be scattered from their presumed near-circular birth orbits to their present-day orbits by secular processes or external perturbations. Two changes in stellar orbits can be decomposed into two different components: the drifting of the angular momentum (ΔL_z , dubbed “radial migration”; Sellwood & Milroy 2002, or “blending”; Schönrich & Milroy 2003), and the contribution to the radial migration or the

radial action (Δr_A , dubbed "radial bending" or "bending").

Many studies have shown that radial migration in the Milky Way disc is strong using age and metallicity of various kind of tracers (e.g. Schönlank & Brainer 2000; Sanders & Brainer 2010; Picaud et al. 2018; Bensado et al. 2021; Xiang et al. 2021; Han et al. 2022a; Heymans et al. 2024). The age and metallicity of stars are important in understanding the migration history of the Galactic disc because they encode the birth radii of stars (Schönlank & Brainer 2000; Minchev et al. 2018; Lu et al. 2023a; Battaglia et al. 2023a). Picaud et al. (2020) quantified the strength of radial migration and heating in the Galactic disc and showed that radial migration in the Galactic disc is significantly more efficient

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than radial heating. Results from analytic studies and simulations have shown that radial migration could reshape the chemical properties of the Galactic disc, such as the azimuthal abundance distribution of the stars (B. Martin et al. 2013). The metallicity-bimodality of the disc

(McMillan 2017). However, in the past, the Galactic bar may not have been rotating with the pattern speed we measure today. It has been demonstrated analytically and computationally that the Galactic bar could decelerate due to dynamical friction against the dark matter



Report Issue

HTML can also make reading papers on your phone easier!



particular the truncation of the high- $[\alpha/\text{Fe}]$ disc beyond the solar circle. Under our proposed radial migration

Universal design:

When spaces or objects are made more accessible (i.e. to people with disabilities), they are made better for *everyone*.

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- Ex., The curb cut

How to work with the HTML format?

Source: arXiv

- Use **packages compatible with LaTeXML**
 - Or ask for packages you use to be made compatible
- Use a standard structure for your papers.
- Use **structures with embedded meaning**
 - Ex., if you want to emphasize text, `\emph{text}` is better than `{\it text}`
- Read HTML papers and **report issues!**



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- Read HTML papers and **report issues!**
- ...and write alternative text



Alternative (alt) text

- **Alternative text** provides textual descriptions of images
 - When a screen reader reaches an image, if there is alt text, that alt text will be read in place of the image.
- Ideally, alt text can replace the image such that a person can understand the content without seeing the image!
- In LaTeX, using **alt from the graphicx** package allows you to include alt text.

```
\begin{figure*}
  \includegraphics[width=6.7in, alt=
    {In the illustrative example of
    BP/RP spectra, the high-[N/O]
    spectrum has much deeper
    absorption features in the CN
    bands than the [N/O]-typical
    spectrum, whereas the carbon-rich
    stellar spectrum has stronger
    absorption still in both the CN,
    CH, and C2 features. The NH band
    and Al lines are not obviously
    visible in the spectra.}]
    {xp_example_v2.png}
```




Credit: Cynthia Bennet

Alternative (alt) text

- **Start with an overview**, then go into detail.
- **Don't repeat information** from the caption.
 - Note that by default, MNRAS seems to repeat the caption as alt text, so you may need to ask them to fix it upon paper publication.
- Focus on **the key information**. For a figure, this is likely to be:
 - Figure type
 - Where the data is coming from
 - Axis labels
 - Key takeaways
- Do not mention purely visual information (i.e., color)



Credit: Cynthia Bennet

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- This is a scenario when **done is better than perfect**.

Alternative (alt) text

Alternative text for *outreach images*?

The JWST team at STScI has done a fantastic job writing beautiful alt text for their public release images.

See the talk at the STScI Day of Accessibility: “**Illuminating the Universe with Accessible Text Descriptions**” by Margaret W. Carruthers



How do we make **conferences** accessible?

- More information is better! Consider adding accessibility information under a “Location” section.
 - Consider **venue accessibility**.
- Use **microphones** for talks.
- Ensure **break times** are not lost in a schedule slip!
- Make sure **seating** is always available.
- **Hybrid conference** formats benefit many!
 - Consider enabling automatic captions.
- In the ideal world, **talk slides should be made available to participants in advance** (and not as PDFs!).



McClurg 2024

**Disability is not
rare in the
sciences.**



“For example, of all employed science, engineering, and health doctorate holders, 5.0% of those under age 40 have a disability, whereas 9.4% of those ages 40 and over have one.”